

The azimuthal dependence of Rayleigh wave ellipticity in a slightly anisotropic medium

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Accepted 2021 January 18. Received 2021 January 15; in original form 2020 April 6

SUMMARY

A recent study analysed the Rayleigh wave ellipticity obtained by ambient noise cross-correlation in periods of 8–20 s, and observed the Rayleigh wave ellipticity is backazimuth-dependent with a 180° periodicity in the contiguous United States. However, the azimuthal anisotropic parameters have not been inverted to depths, and the comparison with other seismic results has not been possible so far, partially due to the lack of related theoretical investigations. Here, we first derive explicit formulation to relate the period-dependent backazimuthal Rayleigh wave ellipticity with the depth-dependent azimuthal wave speed variation in a slightly anisotropic medium based on the variational principle; by carefully examining relations among different parametrizations of a horizontally transverse isotropic medium, we then express the final formulation in terms of Crampin's notation. The formulation is verified by comparison with the results of anisotropic propagator matrix technique. Tests show the backazimuth-dependent Rayleigh wave ellipticity provides complementary information on anisotropic parameters in addition to the widely used phase velocity. A simple application of the derived formulation to real data in North America is also provided. Our formulation can be regarded as an extension of the classic work on azimuthal-dependent phase velocity, and helps to quantitatively explain the backazimuth-dependent Rayleigh wave ellipticity.

Key words: Seismic anisotropy; Surface wave and free oscillations; Theoretical seismology; Wave propagation.

1 INTRODUCTION

Seismic azimuthal anisotropy provides an important constraint on mineral alignment direction, deformation pattern and the Earth's evolution (Maupin & Park 2007). Various seismic data have been used to investigate the azimuthal anisotropy, including the surface wave phase velocity dispersion (Smith & Dahlen 1973; Montagner & Nataf 1986), shear wave splitting (Vinnik et al. 1989; Silver & Chan 1991), Pn azimuthal tomography (Hess 1964), surface wave polarization (Park & Yu 1993; Tanimoto 2004) and teleseismic receiver functions (Bianchi et al. 2010; Liu & Niu 2012). Among these methods, surface waves phase velocity dispersion is widely used and has depth resolution (Smith & Dahlen 1973; Maupin & Park 2007). The surface wave phase velocity dispersion has been used to map azimuthal anisotropy in various areas from the crust to upper mantle (Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a), usually in the framework of the perturbation theory derived from Rayleigh's principle (Smith & Dahlen 1973; Montagner & Nataf 1986).

The perturbation theory builds connection between the backazimuth-dependent Rayleigh wave phase velocity and the depth-dependent anisotropic model parameters, and takes the form

$$\delta c(\theta, z) = \delta c_0(z) + c_1(z) \cos 2\theta + c_2(z) \sin 2\theta + c_3(z) \cos 4\theta + c_4(z) \sin 4\theta, \quad (1)$$

where θ is the azimuth of incident surface wave, and the parameters $c_1 \sim c_4(z)$ are depth-dependent with explicit relation with the model azimuthal anisotropy parameters (Smith & Dahlen 1973). Usually, the 4θ terms of the Rayleigh wave are much smaller than the 2θ terms, and therefore are neglected (Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a). The results of the surface wave phase velocity dispersion perturbation theory are in agreement with the numerical method in a weak anisotropic medium up to 5 percent (Chen & Tromp 2007).

On the other hand, full waveform propagator techniques for a layered anisotropic medium to calculate the phase velocity dispersion curves and eigenfunctions are also available (Anderson 1961; Crampin 1970; Fryer & Frazer 1984; Mallick & Frazer 1990; Park 1996; Thomson 1997). Although accurate results in a strong anisotropic medium can be obtained using the propagator matrix method, the perturbation theory is currently more widely used in inversion (Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a), partially because the perturbation theory is computationally stable, easy to be implemented, and has clear physical meaning.

Since the layered azimuthal anisotropic parameters are still relatively poorly constrained compared with the isotropic shear wave velocity, additional quantities to measure the seismic anisotropic parameters with depth resolution are still needed. Recently, Montagner & Griot-Pommer (2000) and Romanowicz & Yuan (2012) show *SKS* splitting measurements in the presence of layered anisotropy can be modelled using the same parameters of surface wave dispersion; Tanimoto (2004) developed formulation for the surface wave polarization on the Rayleigh–Love mode-coupling theory. However, the *SKS* splitting measurement mainly reflects the anisotropy in the upper mantle, instead of shallower depth such as the crust (Montagner & Griot-Pommer 2000); the utility of the surface wave polarization has been questioned (Maupin 2004), because the coupling effects between the first-overtone Rayleigh wave and fundamental Love wave were neglected in the surface wave polarization theoretical analysis, but may have a large influence in realistic Earth model (Maupin 2004).

In addition to the surface wave phase velocity dispersion, Rayleigh wave ellipticity provides complementary information of the depth-dependent velocity structure, and has been successfully used to invert for the isotropic velocity model (Boore & Nafi Toksöz 1969; Tanimoto & Alvisuri 2006; Tanimoto & Tsuboi 2009). Recently, Rayleigh wave ellipticity obtained from ambient noise cross-correlation has been discovered to be period-dependent (8–20 s) and has the 180° periodicity in back azimuth (Lin & Schmandt 2014). The observed fast directions of Rayleigh wave ellipticity also correlate well with the stress map (Lin & Schmandt 2014). However, the lack of theoretical investigation for the azimuthal-dependent ellipticity on the depth-dependent azimuthal anisotropy parameters restrict the anisotropy parameters to be inverted at depth.

In this study, starting from the analysis of surface wave propagation in a multilayered monoclinic anisotropic medium, we derive partial derivative of Rayleigh wave ellipticity with respect to model parameters (described by the elastic constants C_{mn}) using the variational principle. By carefully examining the different parametrizations of a horizontally transverse isotropic (HTI) medium, we then reveal the relations among components of C_{mn} in an HTI medium with the fast axis in an arbitrary direction within a horizontal plane. Similar to the phase velocity in an azimuthal anisotropic medium, the backazimuth-dependent Rayleigh wave ellipticity is finally expressed as the perturbation on the isotropic parameters (described by Crampin's notation), and takes the similar form as in eq. (1). The formulation is verified by comparison with the results of an anisotropic propagator matrix technique (Park 1996), and can successfully explain previous observed 180° periodicity (Lin & Schmandt 2014). Numerical tests show that Rayleigh wave ellipticity has increased sensitivity to shallower depths compared to surface wave dispersion over the same period. Finally, a simple application of the derived formulation to invert Rayleigh wave ellipticity in North America has been provided. The derived formulation in this study can be regarded as an extension of the classic works on surface wave phase velocity in a slightly anisotropic medium (Smith & Dahlen 1973; Montagner & Nataf 1986), and provides the theoretical foundation to invert the Rayleigh wave ellipticity for the depth-dependent azimuthal anisotropic parameters.

2 THEORY

The Rayleigh wave ellipticity is quantitatively described by the period-dependent horizontal-over-vertical ratio (HV ratio, Lin & Schmandt 2014) or oppositely, vertical-over-horizontal ratio (ZH ratio, Tanimoto & Tsuboi 2009) in literature. Because the ZH ratio is stable even when the vertical displacement disappears within deep basins (Tanimoto & Tsuboi 2009), we use the ZH ratio, hereinafter denoted by ϵ , to quantify the Rayleigh wave ellipticity in this study.

For a multilayered model in the Cartesian coordinates (x_1 – x_2 – x_3) with x_3 -axis in the vertical direction and $x_3 = 0$ at free surface (Fig. 1), the elastic wave equation in each homogeneous layer is described as

$$\rho \frac{\partial^2 u_i}{\partial t^2} = C_{ijkl} \frac{\partial^2 u_l}{\partial x_k \partial x_j}, \quad (2)$$

where C_{ijkl} is the fourth-rank elastic tensor, ρ is the density, and u_i is the i th component of the displacement. Due to the symmetry, C_{ijkl} can be expressed by a 6×6 matrix C_{mn} according to Voigt notation (Maupin & Park 2007).

In this study, instead of the full elastic tensor, it is sufficient to consider a monoclinic anisotropic medium formulated by the general form as (Fryer & Frazer 1987)

$$C_{mn} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & C_{16} \\ C_{12} & C_{22} & C_{23} & 0 & 0 & C_{26} \\ C_{13} & C_{23} & C_{33} & 0 & 0 & C_{36} \\ 0 & 0 & 0 & C_{44} & C_{45} & 0 \\ 0 & 0 & 0 & C_{45} & C_{55} & 0 \\ C_{16} & C_{26} & C_{36} & 0 & 0 & C_{66} \end{pmatrix}, \quad (3)$$

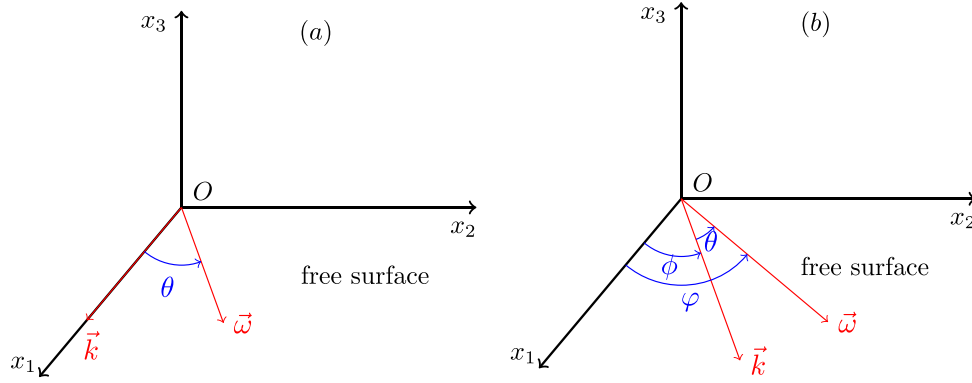


Figure 1. Schematics to show two configurations considered in this study, including the axis of Cartesian coordinates, wave propagation direction, and the fast axis of the HTI media. The x_3 -axis is taken upwardly positive. (a) The incident surface wave propagates along the x_1 -axis described as $\vec{k}$; the fast axis of the medium corresponds to the arbitrary angle within the x_1 - x_2 plane described as $\vec{w}$; the free surface is at $x_3 = 0$; θ is the angle between the $\vec{w}$ and $\vec{k}$. (b) The incident surface wave propagates in direction ϕ relative to x_1 -axis within the x_1 - x_2 plane; and the fast axis of the HTI medium corresponds to the angle φ relative to the x_1 -axis.

which can be used to (but not restricted to) describe the HTI media with arbitrary fast axis direction within $x_1 - x_2$ plane (Fig. 1). Note that in the case of an HTI medium, relations among components of C_{mn} need to be taken into consideration, which will be discussed in detail in Section 2.2.

Without loss of generality, we assume the incident wave propagates along the x_1 -axis (Fig. 1a), and the displacement-stress vector is defined as,

$$\mathbf{y} = (y_1, y_2, y_3, y_4, y_5, y_6)^T, \quad (4)$$

where

$$u_3 = y_1(x_3)e^{-i(\omega t - kx_1)}, \sigma_{33} = y_2(x_3)e^{-i(\omega t - kx_1)}, \quad (5)$$

$$u_1 = iy_3(x_3)e^{-i(\omega t - kx_1)}, \sigma_{31} = iy_4(x_3)e^{-i(\omega t - kx_1)}, \quad (6)$$

$$u_2 = iy_5(x_3)e^{-i(\omega t - kx_1)}, \sigma_{32} = iy_6(x_3)e^{-i(\omega t - kx_1)}, \quad (7)$$

and σ_{3i} is the traction acting on the x_3 -normal plane. The ZH ratio is then described as $\epsilon = \frac{y_1(0)}{y_3(0)}$, where $y_i(0)$ means the value of y_i at the free surface. The second-order wave equation of motion in eq. (2) can be simplified as the first-order ordinary differential equation,

$$\frac{d\mathbf{y}}{dx_3} = \mathbf{M}\mathbf{y}, \quad (8)$$

where

$$\mathbf{M} = \begin{pmatrix} 0 & \frac{1}{C_{33}} & \frac{C_{13}}{C_{33}}k & 0 & \frac{C_{36}}{C_{33}}k & 0 \\ -\rho\omega^2 & 0 & 0 & k & 0 & 0 \\ -k & 0 & 0 & \frac{C_{44}}{C_{44}C_{55}-C_{45}^2} & 0 & -\frac{C_{45}}{C_{55}C_{44}-C_{45}^2} \\ 0 & -\frac{C_{13}}{C_{33}}k & k^2(C_{11}-\frac{C_{13}^2}{C_{33}})-\rho\omega^2 & 0 & k^2(C_{16}-\frac{C_{36}C_{13}}{C_{33}}) & 0 \\ 0 & 0 & 0 & -\frac{C_{45}}{C_{44}C_{55}-C_{45}^2} & 0 & \frac{C_{55}}{C_{44}C_{55}-C_{45}^2} \\ 0 & -\frac{C_{36}}{C_{33}}k & k^2(C_{16}-\frac{C_{13}C_{36}}{C_{33}}) & 0 & (C_{66}-\frac{C_{36}^2}{C_{33}})k^2-\rho\omega^2 & 0 \end{pmatrix}. \quad (9)$$

Note that the number of elastic moduli C_{mn} reduces to 11 in eq. (9) because C_{12} , C_{22} , C_{23} and C_{26} vanish when wave propagation direction is restricted to x_1 (Fig. 1a). The full anisotropic propagator matrix solution can then be obtained using eqs (8) and (9) in a similar approach as in the isotropic medium (Takeuchi & Saito 1972; Hu & Zhu 2017). For the medium of isotropy, vertically transverse isotropy (VTI), and HTI in some special cases ($\vec{w} = \vec{x}_1$ or $\vec{w} = \vec{x}_2$ in Fig. 1), the upper left 4×4 elements ($P - SV$) and lower right 2×2 elements (SH) of $\mathbf{M}$ are decoupled (Takeuchi & Saito 1972), whereas the coupling effect cannot be neglected for general HTI cases. The boundary conditions adopted for the eigenmode are $y_2(0) = y_4(0) = y_6(0) = 0$ at the free surface and radiation boundary condition $y_i(-\infty) = 0$ in the half-space.

Following Tsuboi & Saito (1983) and Tanimoto & Tsuboi (2009), we assume two sets of solutions of eq. (4) y_i ($i = 1 \dots 6$) and z_i ($i = 1 \dots 6$), of which y_i satisfies the free surface and radiation boundary conditions, while z_i only satisfies the radiation boundary condition

in the half-space. Using the explicit expression in eqs (8) and (9), for pseudo-Rayleigh mode, it can be derived,

$$\begin{aligned}
 \frac{d}{dx_3}(z_1 y_2 + z_3 y_4 + z_5 y_6) &= \dot{z}_1 y_2 + z_1 \dot{y}_2 + \dot{z}_3 y_4 + z_3 \dot{y}_4 + \dot{z}_5 y_6 + z_5 \dot{y}_6 \\
 &= (k^2 C_{55} - \rho \omega^2) z_1 y_1 + (k^2 C_{11} - \rho \omega^2) z_3 y_3 + (k^2 C_{66} - \rho \omega^2) z_5 y_5 + \\
 &\quad + C_{33} \dot{z}_1 \dot{y}_1 + C_{55} \dot{z}_3 \dot{y}_3 + C_{44} \dot{z}_5 \dot{y}_5 - \\
 &\quad - k C_{13} (\dot{z}_1 y_3 + z_3 \dot{y}_1) - k C_{36} (\dot{z}_1 y_5 + z_5 \dot{y}_1) + k C_{55} (\dot{z}_3 y_1 + z_1 \dot{y}_3) + \\
 &\quad + k C_{45} (z_1 \dot{y}_5 + \dot{z}_5 y_1) + C_{45} (\dot{z}_3 \dot{y}_5 + \dot{z}_5 \dot{y}_3) + k^2 C_{16} (z_3 y_5 + z_5 y_3) \\
 &\equiv F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega),
 \end{aligned} \tag{10}$$

where $\dot{z}_i$ and $\dot{y}_i$ are the shorthand notations of $\frac{\partial z_i}{\partial x_3}$ and $\frac{\partial y_i}{\partial x_3}$, respectively. The integral of function F over x_3 , denoted by $\int_{-\infty}^0 F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega) dx_3$, can then be regarded as a generalization of the energy integral (Tsuboi & Saito 1983). Therefore, the variational principle similar to Tsuboi & Saito (1983) and Tanimoto & Tsuboi (2009) can be applied to derive the analytic partial derivative of Rayleigh wave ellipticity ϵ with respect to the elastic constant C_{mn} . Note that $F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega)$ is symmetric with respect to $\mathbf{y}$ and $\mathbf{z}$, that is, $F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega)$ remains the same when $\mathbf{y}$ and $\mathbf{z}$ interchange with each other.

Eq. (10) can be applied to calculate the partial derivative of Rayleigh wave ellipticity with respect to model parameters in a general unperturbed monoclinic anisotropic medium. Here, we focus on the special case when the reference model is isotropic or VTI, which is most widely used in practice so far (Smith & Dahlen 1973; Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a). In a weak anisotropic medium, for the isotropic or VTI reference model, the pseudo-Rayleigh and the pseudo-Love modes are approximately decoupled, and y_5 is small compared with y_1 and y_3 in the pseudo-Rayleigh mode displacement vector $\mathbf{y} = (y_1, y_3, y_5)$. In this case, if only the pseudo-Rayleigh wave is considered, to the first order, some of the off-diagonal elastic constants, including C_{36} , C_{45} and C_{16} , are approximately 0. On the other hand, y_5 or z_5 of the pseudo-Rayleigh wave is also a first-order quantity, since it was well observed that the azimuthal anisotropy can rotate the Rayleigh wave polarization ellipse slightly into the transverse-horizontal component and introduce deflection of the Rayleigh wave polarization away from the direct path from source to receiver (Laske & Masters 1998). From a theoretical point of view, the small off-diagonal elastic constants (C_{36} , C_{45} and C_{16}) and small eigenfunctions (z_5 , y_5 , $\dot{z}_5$ and $\dot{y}_5$) are of the same order, because the first-order perturbation of elastic coefficients C_{mn} will introduce first-order perturbation on the eigenfunctions (Tanimoto 2004). The multiplication of two first-order small terms will result in second-order small terms in the $F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega)$ function (eq. 10), including $(k^2 C_{66} - \rho \omega^2) z_5 y_5$, $C_{44} \dot{z}_5 \dot{y}_5$, $C_{36} (\dot{z}_1 y_5 + z_5 \dot{y}_1)$, $k C_{45} (z_1 \dot{y}_5 + \dot{z}_5 y_1)$, $C_{45} (\dot{z}_3 \dot{y}_5 + \dot{z}_5 \dot{y}_3)$ and $k^2 C_{16} (z_3 y_5 + z_5 y_3)$. Therefore, by neglecting the second-order small terms, $F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega)$ for pseudo-Rayleigh wave in a weak anisotropic medium (eq. 10) can be approximately simplified as

$$\begin{aligned}
 F(\mathbf{y}, \mathbf{z}, C_{mn}, \omega) &= C_{11} k^2 z_3 y_3 + C_{55} (k z_1 + \dot{z}_3) (k y_1 + \dot{y}_3) + C_{33} \dot{z}_1 \dot{y}_1 \\
 &\quad - C_{13} k (\dot{z}_1 y_3 + z_3 \dot{y}_1) - \rho \omega^2 (z_1 y_1 + z_3 y_3).
 \end{aligned} \tag{11}$$

The partial derivative of ϵ with respect to C_{11} , C_{13} , C_{33} and C_{55} can then be obtained explicitly using the variational principle (Tsuboi & Saito 1983; Tanimoto & Tsuboi 2009, the explicit expression will be given in Section 2.3); the partial derivatives with respect to all other C_{mn} are approximately zeros. It is noteworthy that the simplification from eq. (10) to eq. (11) is not strict; it requires the weak anisotropic approximation of the medium, which will be validated by numerical tests (see Section 3).

Further theoretical analyses are still in need for the following reasons. First, it is useful to express the resulting formulation in terms of Crampin's notation (see Section 2.1 for detailed explanation), instead of C_{mn} , for comparison with previous studies, since Crampin's notation is usually adopted in previous surface wave phase velocity azimuthal anisotropy studies (Crampin 1981; Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a). Second, because we are most interested in the case of an HTI medium, the relations among C_{mn} in an HTI medium need to be clarified. Third, it is also useful to build connections between the HTI and VTI anisotropies, since the partial derivatives of azimuth-dependent phase velocity in an HTI medium are always calculated by the formulation in a VTI medium in practice (Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a; Liu et al. 2019).

We address these issues in the following sections. We first carefully examine relations among different parametrizations of the anisotropic medium (Sections 2.1 and 2.2). We then reveal the relation of ZH depth-sensitivity kernels in an HTI and VTI medium (Section 2.3). Finally, we derive the corresponding formulation for a general incident surface wave as shown in Fig. 1(b) (Section 2.4).

2.1 Partial derivative of ZH ratio in Crampin's notation

In a slightly anisotropic medium, another parametrization, other than C_{mn} , exists to express the anisotropy, including the five background VTI parameters ($\bar{A}$, $\bar{C}$, $\bar{F}$, $\bar{L}$, $\bar{N}$), plus eight azimuthal anisotropic parameters (B_c , B_s , G_c , G_s , H_c , H_s , C_c , C_s), which is widely used in body and surface wave anisotropy studies in regional and global scales (Crampin 1981; Montagner & Nataf 1986; Yao et al. 2010). The total 13 parameters, expressed in Crampin's notations as suggested in Montagner & Nataf (1986), can be derived as linear combinations of the elastic constants C_{mn} . The parameters in Crampin's notation were explicitly expressed as the function of C_{mn} (Montagner & Nataf 1986, eqs 5a–c). Therefore, with the chain rule, the partial derivative of ZH ratio ϵ with respect to parameters in C_{mn} and Crampin's notations have the following relations,

$$\frac{\partial \epsilon}{\partial B_c} = \frac{\partial \epsilon}{\partial C_{11}} - \frac{\partial \epsilon}{\partial C_{22}}, \quad (12)$$

$$\frac{\partial \epsilon}{\partial B_s} = \frac{1}{2} \frac{\partial \epsilon}{\partial C_{16}} + \frac{1}{2} \frac{\partial \epsilon}{\partial C_{26}}, \quad (13)$$

$$\frac{\partial \epsilon}{\partial G_c} = \frac{\partial \epsilon}{\partial C_{55}} - \frac{\partial \epsilon}{\partial C_{44}}, \quad (14)$$

$$\frac{\partial \epsilon}{\partial G_s} = \frac{\partial \epsilon}{\partial C_{45}}, \quad (15)$$

$$\frac{\partial \epsilon}{\partial H_c} = \frac{\partial \epsilon}{\partial C_{13}} - \frac{\partial \epsilon}{\partial C_{23}}, \quad (16)$$

$$\frac{\partial \epsilon}{\partial H_s} = \frac{\partial \epsilon}{\partial C_{36}}. \quad (17)$$

It is noteworthy that the above equations assume that each component of C_{mn} is independent from each other, which is, however, not the case for an HTI medium considered in this study.

2.2 Partial derivative of ZH ratio in Backus's notation

We also use the following formulae to parametrize an HTI medium with weak anisotropy (Backus 1965; Crampin 1981; Park 1996),

$$\rho \alpha^2(\theta) = A + B \cos 2\theta + C \cos 4\theta, \quad (18)$$

$$\rho \beta^2(\theta) = D + E \cos 2\theta, \quad (19)$$

where α and β are the azimuth-dependent compressional and shear wave velocities in an HTI medium, θ is the angle between wave propagation and the fast axis (see Fig. 1). In an isotropic medium, $(B, C, E) = (0, 0, 0)$, $A = \lambda + 2\mu$, and $D = \mu$. We hereinafter refer to the parametrization (A, B, C, D, E) as Backus's notation.

The relation between the elastic coefficients C_{mn} and (A, B, C, D, E) was derived by Park (1996) in Cartesian coordinates in implicit form, and is given in explicit form as follows,

$$C_{11} = A + B \cos 2\theta + C \cos 4\theta, \quad (20)$$

$$C_{22} = A - B \cos 2\theta + C \cos 4\theta, \quad (21)$$

$$C_{33} = A - B + C, \quad (22)$$

$$C_{44} = D - E \cos 2\theta, \quad (23)$$

$$C_{55} = D + E \cos 2\theta, \quad (24)$$

$$C_{66} = C + D + E - C \cos 4\theta, \quad (25)$$

$$C_{12} = A - 2C - 2D - 2E - C \cos 4\theta, \quad (26)$$

$$C_{13} = A - \frac{1}{2}B - C - 2D + \left(\frac{1}{2}B - 2C - 2E\right) \cos 2\theta, \quad (27)$$

$$C_{16} = \frac{1}{2}B \sin 2\theta + C \sin 4\theta, \quad (28)$$

$$C_{26} = \frac{1}{2}B \sin 2\theta - C \sin 4\theta, \quad (29)$$

$$C_{36} = \left(\frac{1}{2}B - 2C - 2E\right) \sin 2\theta, \quad (30)$$

$$C_{23} = A - \frac{1}{2}B - C - 2D - \left(\frac{1}{2}B - 2C - 2E\right) \cos 2\theta, \quad (31)$$

$$C_{45} = E \sin 2\theta. \quad (32)$$

Note that eqs (20)–(32) are valid for a general incident surface wave within the $x_1 - x_2$ plane (Fig. 1b).

Combining the above relation between C_{mn} and parameters in Backus's notation with the relation between parameters in Crampin's notation and C_{mn} , we have

$$B_c = B \cos 2\theta, \quad (33)$$

$$B_s = B \sin 2\theta, \quad (34)$$

$$G_c = E \cos 2\theta, \quad (35)$$

$$G_s = E \sin 2\theta, \quad (36)$$

$$H_c = \left(\frac{1}{2} B - 2C - 2E \right) \cos 2\theta, \quad (37)$$

$$H_s = \left(\frac{1}{2} B - 2C - 2E \right) \sin 2\theta, \quad (38)$$

$$C_c = C \cos 4\theta, \quad (39)$$

$$C_s = C \sin 4\theta. \quad (40)$$

Because the 4θ terms are usually at least first-order weaker than the 2θ terms (i.e. $C \ll B$ and $C \ll E$), and are difficult to be measured in reality (Montagner & Nataf 1986; Yao et al. 2010; Lin et al. 2010a), we only consider the terms involving 2θ in the following discussion.

For comparison with Park's (1996) results, from eqs (33)–(40), we arrive at,

$$\frac{\partial \epsilon}{\partial B} = \left(\frac{\partial \epsilon}{\partial B_c} + \frac{1}{2} \frac{\partial \epsilon}{\partial H_{c,s}} \right) \cos 2\theta + \left(\frac{\partial \epsilon}{\partial B_s} + \frac{1}{2} \frac{\partial \epsilon}{\partial H_{c,s}} \right) \sin 2\theta, \quad (41)$$

$$\frac{\partial \epsilon}{\partial E} = \left(\frac{\partial \epsilon}{\partial G_c} - 2 \frac{\partial \epsilon}{\partial H_{c,s}} \right) \cos 2\theta + \left(\frac{\partial \epsilon}{\partial G_s} - 2 \frac{\partial \epsilon}{\partial H_{c,s}} \right) \sin 2\theta. \quad (42)$$

Therefore, we can define

$$\left(\frac{\partial \epsilon}{\partial B_{c,s}} \right)_{HTI} = \frac{\partial \epsilon}{\partial B_{c,s}} + \frac{1}{2} \frac{\partial \epsilon}{\partial H_{c,s}}, \quad (43)$$

$$\left(\frac{\partial \epsilon}{\partial G_{c,s}} \right)_{HTI} = \frac{\partial \epsilon}{\partial G_{c,s}} - 2 \frac{\partial \epsilon}{\partial H_{c,s}}. \quad (44)$$

The subscript *HTI* in the above equations means the corresponding partial derivatives are valid in an HTI medium. Eqs (43)–(44) reveal the relations satisfied by parameters in Crampin's notation; since the parameters in Crampin's notation and elastic constants have explicit relations (Montagner & Nataf 1986), eqs (43)–(44) also reveal the relations satisfied by parameters in elastic constants C_{mn} in an HTI medium.

2.3 Relation with the depth-sensitivity kernel in a VTI medium

As derived in the preceding sections, the explicit formulation of the analytic partial derivative $\left(\frac{\partial \epsilon}{\partial B_{c,s}} \right)_{HTI}$ and $\left(\frac{\partial \epsilon}{\partial G_{c,s}} \right)_{HTI}$ can be calculated using eqs (12)–(17) and eqs (43)–(44).

In previous surface wave phase velocity azimuthal anisotropy studies, the partial derivative of azimuth-dependent phase velocity in an HTI medium is always calculated by the formulation in a VTI medium in practice (Yao et al. 2010; Lin et al. 2010a). In a similar way, as follows, we show the partial derivative of Rayleigh wave ellipticity can also be calculated by relating to the depth-sensitivity kernel of a VTI medium. Because the Rayleigh wave ellipticity and its sensitivity kernel with respect to model parameters in a VTI medium have been well implemented both in Cartesian (Herrmann 2013) and spherical earth (Masters et al. 2014) coordinate systems, the partial derivative in a slightly HTI medium can therefore be stably obtained.

We consider a VTI medium parameterized with the Love notation ($\tilde{A}$, $\tilde{C}$, $\tilde{F}$, $\tilde{L}$, $\tilde{N}$) (Maupin & Park 2007). The depth-sensitivity kernel of the VTI medium is relatively straightforward, as it is similar to an isotropic medium using the variational principle (see Appendix A). In the derivation, the key step is to obtain the corresponding quantity $F(\mathbf{y}, \mathbf{z}, \tilde{A}, \tilde{C}, \tilde{F}, \tilde{L}, \omega)$. The F function for a VTI medium is (for details, see Appendix A, eq. A12)

$$\begin{aligned} F(\mathbf{y}, \mathbf{z}, \tilde{A}, \tilde{C}, \tilde{L}, \tilde{F}, \omega) &\equiv \frac{d}{dz} (z_1 y_2 + z_3 y_4) = \\ &\tilde{A} k^2 z_3 y_3 + \tilde{C} \dot{z}_1 \dot{y}_1 + \tilde{L} (k z_1 + \dot{z}_3) (k y_1 + \dot{y}_3) - \tilde{F} k (\dot{z}_1 y_3 + z_3 \dot{y}_1) \\ &- \rho \omega^2 (z_1 y_1 + z_3 y_3). \end{aligned} \quad (45)$$

By comparing the formulation of the F function in a VTI medium (eq. 45) and an HTI medium (eq. 11), the following relation can be established,

$$\frac{\partial \epsilon}{\partial C_{11}} = \frac{\partial \epsilon}{\partial \tilde{A}}, \quad (46)$$

$$\frac{\partial \epsilon}{\partial C_{33}} = \frac{\partial \epsilon}{\partial \tilde{C}}, \quad (47)$$

$$\frac{\partial \epsilon}{\partial C_{55}} = \frac{\partial \epsilon}{\partial \tilde{L}}, \quad (48)$$

$$\frac{\partial \epsilon}{\partial C_{13}} = \frac{\partial \epsilon}{\partial \tilde{F}}, \quad (49)$$

and all other components of $\frac{\partial \epsilon}{\partial C_{mn}}$ are approximately zeros in the case we consider.

Therefore, with eqs (12)–(17) and eqs (46)–(49), eqs (41) and (42) are simplified as,

$$\frac{\partial \epsilon}{\partial B} = \left(\frac{\partial \epsilon}{\partial \tilde{A}} + \frac{1}{2} \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos 2\theta, \quad (50)$$

$$\frac{\partial \epsilon}{\partial E} = \left(\frac{\partial \epsilon}{\partial \tilde{L}} - 2 \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos 2\theta, \quad (51)$$

which draw connections between HTI anisotropy on the left-hand side with the VTI anisotropy on the right-hand side of the equations. The above equations determine how the $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ vary with azimuth according to the fast axis direction θ when the incident wave propagates along the x_1 -axis (Fig. 1a).

2.4 Formulation for the general incident surface wave

For a general case when the wave propagates in a angle of ϕ from x_1 -axis (Fig. 1b), the formulation in eqs (50) and (51) can be generalized by setting $\theta = \varphi - \phi$ as,

$$\begin{aligned} \frac{\partial \epsilon}{\partial B} &= \left(\frac{\partial \epsilon}{\partial \tilde{A}} + \frac{1}{2} \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos (2\varphi - 2\phi) \\ &= \left[\left(\frac{\partial \epsilon}{\partial \tilde{A}} + \frac{1}{2} \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos 2\phi \right] \cos 2\varphi + \left[\left(\frac{\partial \epsilon}{\partial \tilde{A}} + \frac{1}{2} \frac{\partial \epsilon}{\partial \tilde{F}} \right) \sin 2\phi \right] \sin 2\varphi. \end{aligned} \quad (52)$$

and

$$\begin{aligned} \frac{\partial \epsilon}{\partial E} &= \left(\frac{\partial \epsilon}{\partial \tilde{L}} - 2 \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos (2\varphi - 2\phi) \\ &= \left[\left(\frac{\partial \epsilon}{\partial \tilde{L}} - 2 \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos 2\phi \right] \cos 2\varphi + \left[\left(\frac{\partial \epsilon}{\partial \tilde{L}} - 2 \frac{\partial \epsilon}{\partial \tilde{F}} \right) \sin 2\phi \right] \sin 2\varphi. \end{aligned} \quad (53)$$

Eqs (43) and (44) can therefore be generalized to be

$$\left(\frac{\partial \epsilon}{\partial B_c} \right)_{\text{HTI}}^{\phi} = \left(\frac{\partial \epsilon}{\partial \tilde{A}} + \frac{1}{2} \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos 2\phi, \quad (54)$$

$$\left(\frac{\partial \epsilon}{\partial B_s} \right)_{\text{HTI}}^{\phi} = \left(\frac{\partial \epsilon}{\partial \tilde{A}} + \frac{1}{2} \frac{\partial \epsilon}{\partial \tilde{F}} \right) \sin 2\phi, \quad (55)$$

$$\left(\frac{\partial \epsilon}{\partial G_c} \right)_{\text{HTI}}^{\phi} = \left(\frac{\partial \epsilon}{\partial \tilde{L}} - 2 \frac{\partial \epsilon}{\partial \tilde{F}} \right) \cos 2\phi, \quad (56)$$

$$\left(\frac{\partial \epsilon}{\partial G_s} \right)_{\text{HTI}}^{\phi} = \left(\frac{\partial \epsilon}{\partial \tilde{L}} - 2 \frac{\partial \epsilon}{\partial \tilde{F}} \right) \sin 2\phi, \quad (57)$$

where the ϕ is the direction of incident wave propagation and φ is the fast axis of the HTI medium within the $x_1 - x_2$ plane (Fig. 1b). The above formulation (eqs 54–57) can be readily used to constrain layered azimuthal anisotropy for an isotropic or VTI reference model. The superscript φ and subscript *HTI* will be dropped in the following texts for simplicity without causing confusion.

3 NUMERICAL VERIFICATION

We test the derived formulae in the preceding sections by comparing the results with the anisotropic propagator matrix method of Park (1996), which accurately computes eigenvalues and eigenfunctions in a slightly anisotropic multilayered medium. The anisotropic propagator matrix method, implemented by the FORTRAN package *aniprop* (Park 1996), uses complex numbers to describe the fundamental eigenfunctions to take into consideration of the media attenuation. Therefore, we compute the Rayleigh wave ellipticity using *aniprop* as,

$$\epsilon = \frac{|y_3(0)|}{|y_1(0)|}, \quad (58)$$

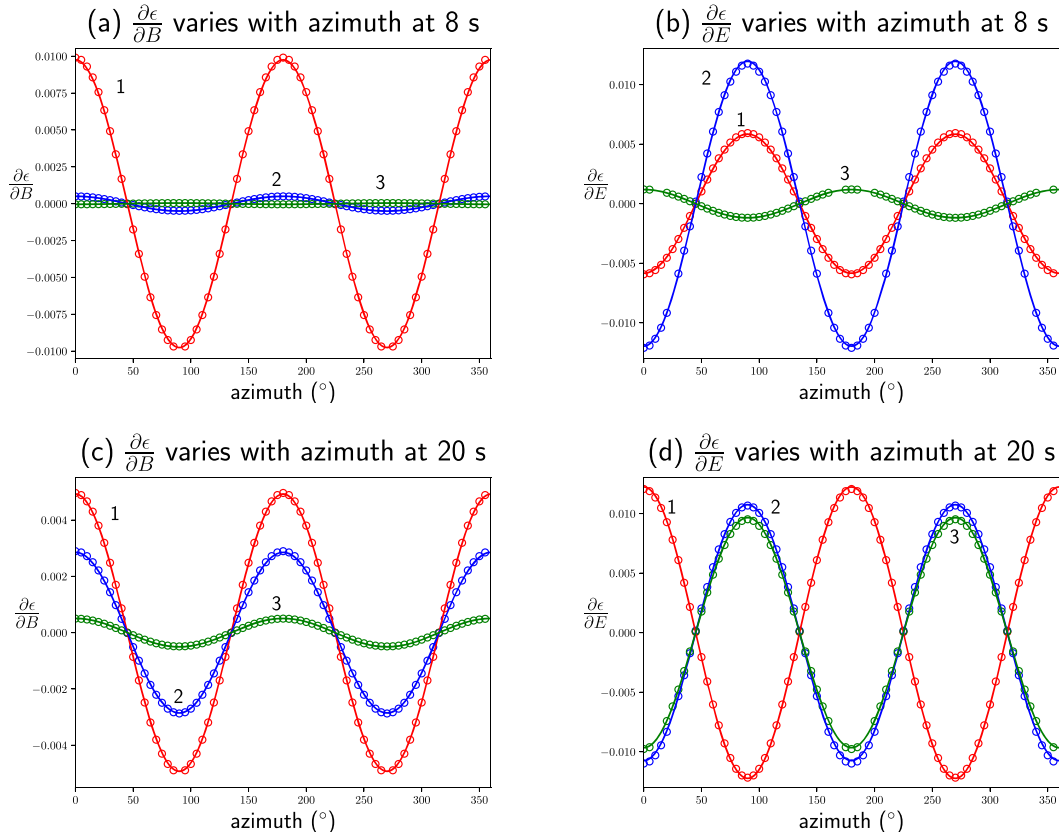


Figure 2. Comparison of how $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ vary with azimuths between finite-difference method and analytic formulation in eqs (50) and (51) at 8 and 20 s. The circles denote the values obtained by using *aniprop* with finite-difference method (perturbation size 1 per cent); the lines denote the values obtained by eqs (50) and (51). (a) $\frac{\partial \epsilon}{\partial B}$ varies with azimuth at 8 s; (b) $\frac{\partial \epsilon}{\partial E}$ varies with azimuth at 8 s; (c) $\frac{\partial \epsilon}{\partial B}$ varies with azimuth at 20 s and (d) $\frac{\partial \epsilon}{\partial E}$ varies with azimuth at 20 s. The i th layer is labelled next to each line ($i=1,2,3$; see Table 1). The colours red, blue and green denote the partial derivatives of ZH ratio with respect to model parameters in the first, second and third layers, respectively.

where the $|y_3(0)|$ and $|y_1(0)|$ denote the modulus of the fundamental eigenfunctions in the vertical and radial directions at the free surface, respectively. In the *aniprop* code, the incident wave is assumed to be along the x_1 -axis, so we choose to validate the key formulae in eqs (50) and (51), instead of eqs (54)–(57).

The formulae in eqs (50)–(51) are tested in the 1-D Hadley–Kanamori velocity model of southern California (H-K model, table 1; Hadley & Kanamori 1977), which contains three layers over a half-space. We first compute the Rayleigh wave ZH ratio over periods of 8–20 s using the *aniprop* package; we then perturb B and E in each layer of the H-K model by 1 per cent, and calculate the $\frac{\Delta \epsilon}{\Delta B}$ and $\frac{\Delta \epsilon}{\Delta E}$ for the models with the fast axis in directions of 0° – 360° using the finite-difference method. In the meantime, the $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ are also computed using eqs (50)–(51), what we referred to as analytic formulation. The Rayleigh wave ellipticity measurements of the reference model were obtained using the *Computer Programs in Seismology* (CPS) package (Herrmann 2013). Finally, the $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ obtained by the finite-difference method and eqs (50)–(51) are compared in different azimuthal angles 0° – 360° at 8 and 20 s (Fig. 2). Also, the peak amplitude of the period-dependent quantities $(\frac{\partial \epsilon}{\partial A} + \frac{1}{2} \frac{\partial \epsilon}{\partial F})$ and $(\frac{\partial \epsilon}{\partial L} - 2 \frac{\partial \epsilon}{\partial F})$ are compared with the finite-difference method (Fig. 3).

These comparisons show that the $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ obtained by the finite-difference method and eqs (50) and (51) agree with each other in the periods of 8–20 s in a slightly anisotropic medium with different fast axis directions, therefore validating our derivation and formulations.

4 DISCUSSION AND APPLICATION

4.1 Rayleigh wave ellipticity as a complementary constraint on azimuthal anisotropy

In an isotropic medium, the depth sensitivity of the Rayleigh wave ellipticity and Rayleigh wave phase velocity dispersion are different at the same period, and so they provide complementary information when jointly inverted for Earth velocity structure (Lin et al. 2010b). In a further test, we calculated the azimuth-dependent depth-sensitivity kernels of the Rayleigh wave ZH ratio with respect to $\tilde{A}$, $\tilde{F}$, $\tilde{L}$ (Figs 4a and b); these partial derivatives were also combined as $(\frac{\partial \epsilon}{\partial A} + \frac{1}{2} \frac{\partial \epsilon}{\partial F})$ and $(\frac{\partial \epsilon}{\partial L} - 2 \frac{\partial \epsilon}{\partial F})$ to represent the depth-dependent azimuthal anisotropic properties (Figs 4c and d). The corresponding phase velocity kernels were also calculated (Figs 4e and f) at 8 and 20 s; the depth kernels of ZH ratio were then compared with the corresponding kernels of phase velocity at the same period (Fig. 4). These comparisons show that the

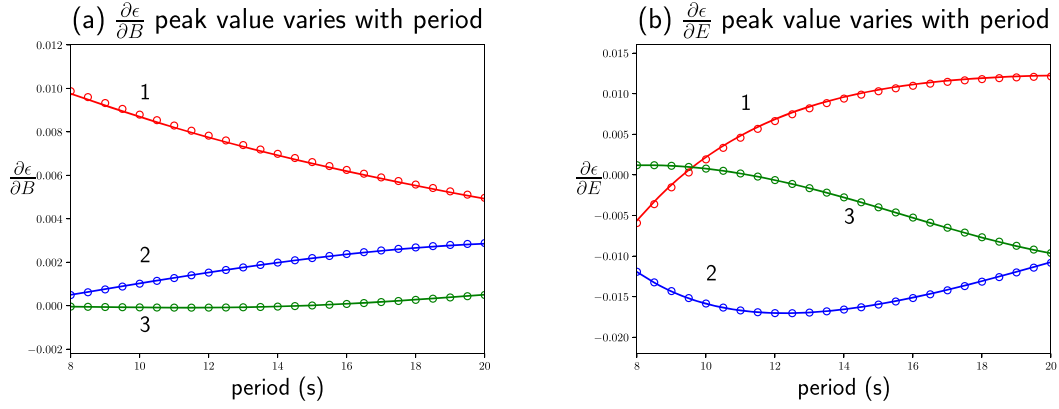


Figure 3. Numerical comparison of how peak amplitude values of $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ varies with periods (8–20 s) between the finite-difference method and analytic formulation in eqs (50) and (51). The sign of $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ peak values is the same as the sign of the corresponding values at azimuth 0° in Fig. 2. The circles denote the azimuthal amplitude of the peak values of $\frac{\partial \epsilon}{\partial B}$ and $\frac{\partial \epsilon}{\partial E}$ computed by finite-difference method using *aniprop*; the lines are the corresponding values obtained by eqs (50) and (51). (a) How peak value of $\frac{\partial \epsilon}{\partial B}$ varies with period; and (b) how peak value of $\frac{\partial \epsilon}{\partial E}$ varies with period. The i th layer is labelled next to each line ($i=1,2,3$; see Table 1). The colours red, blue and green denote the partial derivative of ZH ratio with respect to model parameters in the first, second and third layers, respectively.

Table 1. The H-K velocity model.

Th. (km)	β (km s $^{-1}$)	α (km s $^{-1}$)	ρ (g cm $^{-3}$)	Q_β	Q_α
5.5	3.18	5.50	2.53	600	1200
10.5	3.64	6.30	2.79	600	1200
16.0	3.87	6.70	2.91	600	1200
	4.50	7.80	3.27	900	1800

sensitivity depth of the backazimuth-dependent Rayleigh wave phase velocity dispersion is deeper than Rayleigh wave ellipticity at the same period.

We carry out several synthetic tests to demonstrate that the backazimuth-dependent Rayleigh wave ellipticity can provide a complementary constraint on the depth-dependent azimuthal anisotropic parameters (Fig. 5). We first build up the formulae for forward simulation and inversion as follows.

The perturbation of ZH ratio due to azimuthal anisotropy at period T_i can be formulated, following eqs (43) and (44), as

$$\delta \epsilon(T_i) = \sum_j \left(\frac{\partial \epsilon(T_i)}{\partial G_c(h_j)} G_c(h_j) + \frac{\partial \epsilon(T_i)}{\partial G_s(h_j)} G_s(h_j) + \frac{\partial \epsilon(T_i)}{\partial B_c(h_j)} B_c(h_j) + \frac{\partial \epsilon(T_i)}{\partial B_s(h_j)} B_s(h_j) \right), \quad (59)$$

where h_j is the j th layer of the model. As a proof of concept, we assume $B_{c,s}/\tilde{A} = G_{c,s}/\tilde{L}$ at each layer in this study for simplicity. Therefore, we formulate the problem for all periods as

$$\begin{bmatrix} \frac{\partial \epsilon(T_1)}{\partial \hat{G}_c(h_1)} & \frac{\partial \epsilon(T_1)}{\partial \hat{G}_s(h_1)} & \cdots & \frac{\partial \epsilon(T_1)}{\partial \hat{G}_c(h_m)} & \frac{\partial \epsilon(T_1)}{\partial \hat{G}_s(h_m)} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\partial \epsilon(T_n)}{\partial \hat{G}_c(h_1)} & \frac{\partial \epsilon(T_n)}{\partial \hat{G}_s(h_1)} & \cdots & \frac{\partial \epsilon(T_n)}{\partial \hat{G}_c(h_m)} & \frac{\partial \epsilon(T_n)}{\partial \hat{G}_s(h_m)} \end{bmatrix} \begin{bmatrix} \hat{G}_c(h_1) \\ \hat{G}_s(h_1) \\ \vdots \\ \hat{G}_c(h_m) \\ \hat{G}_s(h_m) \end{bmatrix} = \begin{bmatrix} \delta \epsilon(T_1) \\ \vdots \\ \delta \epsilon(T_n) \end{bmatrix}, \quad (60)$$

where $\hat{G}_{c,s}$ is the normalized azimuthal parameter, that is, $\hat{G}_{c,s} = \frac{G_{c,s}}{\tilde{L}} = \frac{B_{c,s}}{\tilde{A}}$; the depth-sensitivity kernel is calculated as $\frac{\partial \epsilon}{\partial \hat{G}_{c,s}} = \tilde{L} \frac{\partial \epsilon}{\partial B_{c,s}} + \tilde{A} \frac{\partial \epsilon}{\partial G_{c,s}}$. Note that because the $\frac{\partial \epsilon}{\partial \tilde{F}}$ term is included in our kernel computation, the assumption $B_{c,s}/\tilde{A} = G_{c,s}/\tilde{L}$ is different from previously adopted (Lin et al. 2010a; Yao 2015; Liu et al. 2019). Instead, according to eqs (52) and (53), this assumption is equivalent to $B/A = E/D$ in Backus's notation. The more realistic assumptions on the relation between B/A and E/D can be made according to specific circumstances (see numerical simulations in Park 1996, e.g.). The linearized problem is solved by a standard least square approach, with a first-order smoothing constraint between adjacent layers. According to eqs (54)–(57) and Fig. 2, the ZH ratio perturbation, described by $\hat{G}_{c,s}$ in eq. (60), is backazimuth-dependent with a 180° periodicity in a slightly HTI medium. Therefore, after the depth-dependent $\hat{G}_{c,s}$ are obtained, the resulting azimuthal anisotropic amplitude and direction are calculated as $A = \sqrt{\hat{G}_c^2 + \hat{G}_s^2}$ and $\varphi = \frac{1}{2} \arctan \left(\frac{\hat{G}_s}{\hat{G}_c} \right)$, respectively.

In the synthetic tests, we used the H-K model as the ground-truth velocity model, and added azimuthal anisotropy to the H-K model above 50 km depth, where the amplitude of the anisotropy was set to be 5 per cent (for both $B_{c,s}$ and $G_{c,s}$) and the direction of the anisotropy varied according to the index of each layer (Fig. 5a). The synthetic backazimuth-dependent ZH ratio was calculated using eq. (60) over periods of 8–20 s. The synthetic phase wave velocity in the period band 8–40 s was also computed using a similar formula as eq. (60), but without $\frac{\partial c}{\partial \tilde{F}}$ in the kernel computation (Montagner & Nataf 1986). Note that we did not add noise to the synthetic data. During the inversion, we used

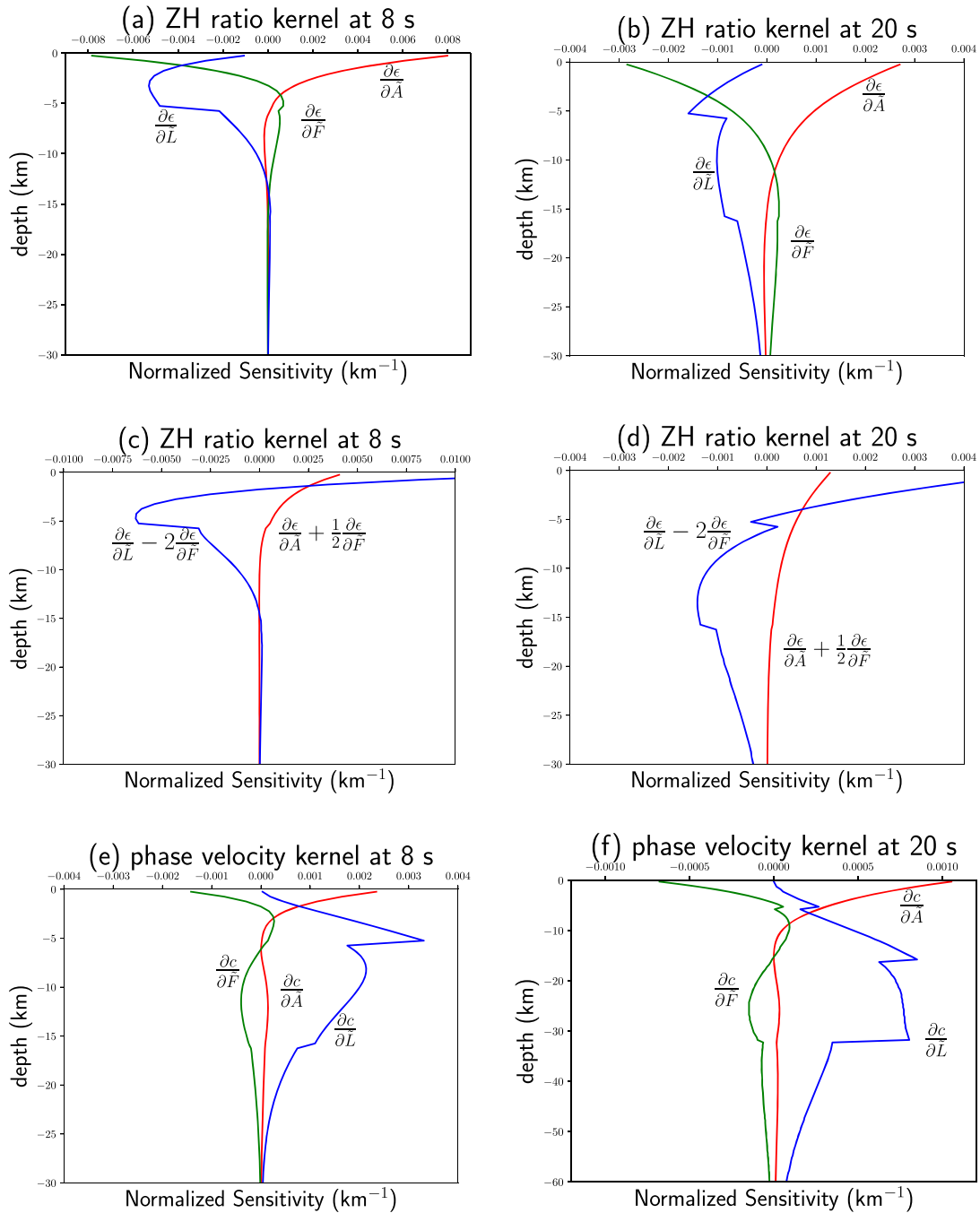


Figure 4. Comparison of the depth kernel between Rayleigh wave ellipticity and Rayleigh wave phase velocity dispersion at 8 and 20 s in a VTI medium. All coloured lines are labelled. (a) The depth kernel of ZH ratio with respect to $\tilde{A}$, $\tilde{L}$ and $\tilde{F}$ at 8 s. (b) The depth kernel of ZH ratio with respect to $\tilde{A}$, $\tilde{L}$ and $\tilde{F}$ at 20 s. (c) The combined depth kernel of ZH ratio calculated by eq. (50) at 8 s. (d) The combined depth kernel of ZH ratio calculated by eq. (50) at 20 s. (e) The depth kernel of Rayleigh wave phase velocity with respect to $\tilde{A}$, $\tilde{L}$ and $\tilde{F}$ at 8 s. (f) The depth kernel of Rayleigh wave phase velocity with respect to $\tilde{A}$, $\tilde{L}$ and $\tilde{F}$ at 20 s.

the ground-truth velocity model as the reference model, and only inverted for the anisotropic parameters. We performed the inversion several times using different synthetic data sets, including the phase velocity alone (Fig. 5b), the ZH ratio alone (Fig. 5c) and the phase velocity and ZH ratio jointly (Fig. 5d). In the joint inversion, the weights for the phase velocity and ZH ratio were set to be equal. Fig. 5 shows the ground-truth velocity (red lines), the anisotropy models (blue bars) and the corresponding synthetic inversion results (green bars), where the length of the bars gives the amplitude of the anisotropy and the direction of the bars gives the azimuth of the fast polarization axis (north pointing upward and east pointing eastward). The synthetic tests indicate that inversion using the phase velocity (8–40 s) or ZH ratio (8–20 s) alone cannot recover the amplitude of the anisotropic parameters exactly. Specifically, the anisotropic parameters in the shallow crust (above 15 km depth) are relatively poorly constrained by the phase velocity alone, whereas the ZH ratio is not sensitive to the anisotropic structure

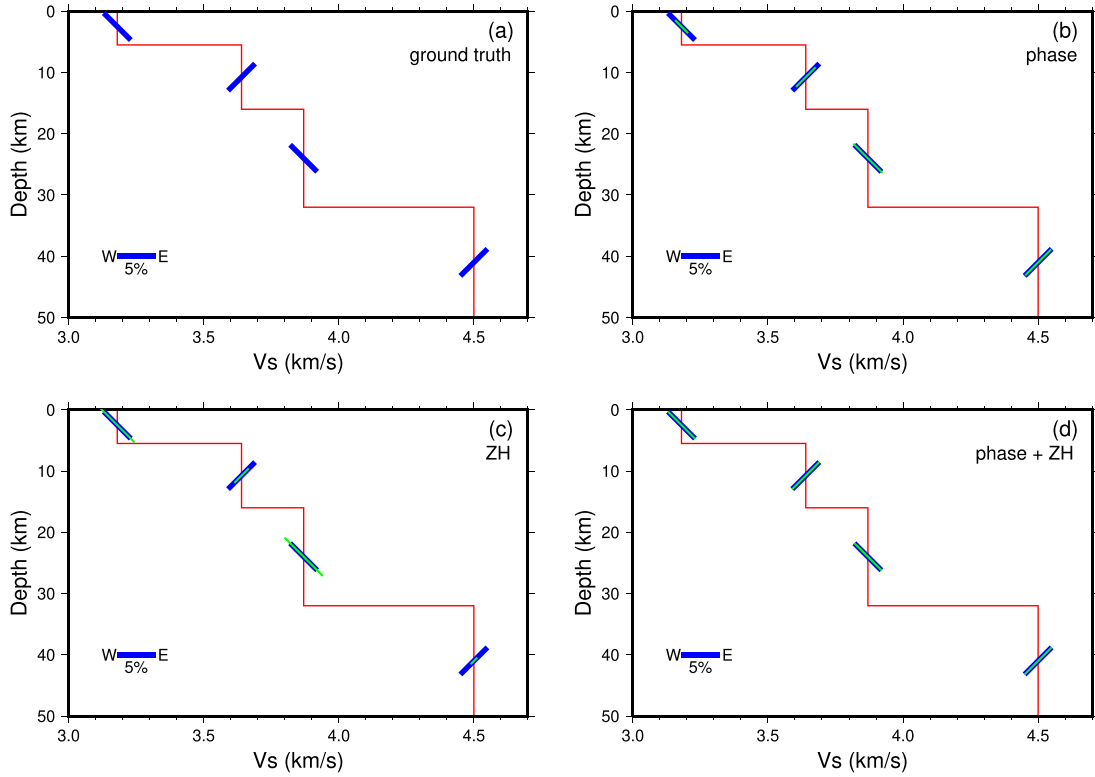


Figure 5. Azimuthal anisotropy synthetic inversion tests using phase velocity (8–40 s), Rayleigh wave ellipticity (8–20 s) separately (b) and (c), and jointly (d). The velocity of the ground-truth model is the same as the H-K model (red lines). The amplitude of the bars denote the amplitude of the azimuthal anisotropy. The direction of the bars denote the azimuth of the fast polarization axis (north pointing upward and east pointing eastward). The amplitude of the ground-truth azimuthal anisotropy is set to be 5 per cent, and the direction of the ground-truth azimuthal anisotropy varies at each layer (blue bars). Only the azimuthal anisotropic parameters are inverted for in the tests, and the ground-truth velocity model is used for kernel computation. The green bars in (b)–(d) denote the inverted azimuthal anisotropy models using different data sets. No noise was added to the synthetic data.

below 30 km. On the contrary, the joint inversion of phase velocity and ZH ratio can exactly recover the anisotropic parameters down to the depth of 50 km.

It is noteworthy that although the formulations of depth-dependent kernels of ZH ratio and Rayleigh wave phase velocities for a slightly anisotropic medium look similar, the term $\frac{\partial \epsilon}{\partial F}$ is involved in the ZH ratio formulation, whereas this term was ignored in the surface wave phase velocity dispersion studies in practice (Yao et al. 2010; Lin et al. 2010a). It is reasonable because $\frac{\partial \epsilon}{\partial F}$ is of the same order as $\frac{\partial \epsilon}{\partial A}$ and $\frac{\partial \epsilon}{\partial L}$ at shallower depths, but as noted previously (Montagner & Nataf 1986), $\frac{\partial \epsilon}{\partial F}$ seems much smaller than $\frac{\partial \epsilon}{\partial L}$ (Figs 4e and f) in the crustal scale.

In phase velocity anisotropy studies, it was previously found that the backazimuth-dependent phase velocity has higher sensitivity with respect to $G_{c,s}$ than $B_{c,s}$ in the middle/lower crust and uppermost mantle, but is more sensitive to $B_{c,s}$ than $G_{c,s}$ at upper crust (Lin et al. 2010a). This is also observed in our numerical simulation for the azimuthal anisotropic phase velocity (Figs 4e and f). However, in our numerical tests, it seems that the Rayleigh wave ellipticity at a particular period is always more sensitive to $G_{c,s}$ than $B_{c,s}$ in all depths (Figs 4c and d), which is different from the phase velocity. This difference is partially due to the fact that the $\frac{\partial \epsilon}{\partial F}$ term is included in our kernel computation in the HTI medium (see $\frac{\partial \epsilon}{\partial A}$ and $\frac{\partial \epsilon}{\partial L}$ in Figs 4a and b for comparison).

To demonstrate the effects of the $\frac{\partial \epsilon}{\partial F}$ term on the inversion, we repeat the synthetic test using ZH ratio alone (Fig. 5c) using synthetic ZH ratio data over periods of 8–50 s. The amplitude of azimuthal anisotropy in the ground-truth model is 5 per cent (Fig. 5a). In this test, when the $\frac{\partial \epsilon}{\partial F}$ term is included in the kernel computation, the ZH ratio in 8–50 s can constrain the anisotropic parameters down to the depth of 50 km (Fig. 6a). However, when the $\frac{\partial \epsilon}{\partial F}$ term is ignored in the ZH ratio kernel computation (Fig. 6b), the recovered amplitude of the anisotropy is approximately 6 per cent, indicating that the amplitude of the azimuthal anisotropic parameters will be overestimated as large as approximately 20 per cent with incorrect ZH ratio kernels.

4.2 Application to field data in North America

We further provide a simple application of our formulation to invert the observed backazimuth-dependent Rayleigh wave ZH ratio for depth-dependent azimuthal parameters located at $(-113^\circ, 42^\circ)$ in western America. The period-dependent Rayleigh wave ellipticity data were obtained by Lin & Schmandt (2014) from noise cross-correlation functions.

During the inversion, we chose the V_s model obtained by Schmandt et al. (2015) (Figs 7a and c), combined with a constant V_p/V_s ratio of 1.73, as the reference model to compute the depth-sensitivity kernel of the ZH ratio. The simple assumption $B_{c,s}/\tilde{A} = G_{c,s}/\tilde{L}$ was used.

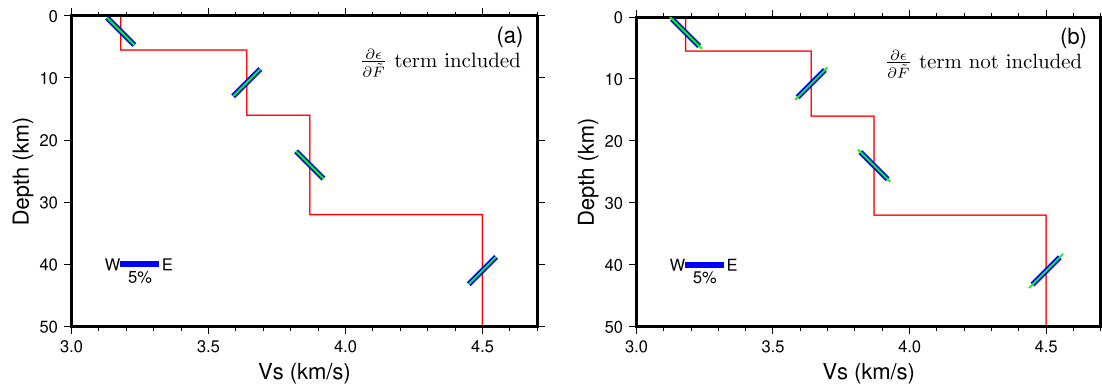


Figure 6. Azimuthal anisotropy inversion synthetic tests using backazimuth-dependent ZH ratio (8–50 s) using the depth kernels with (a) and without (b) $\frac{\partial \epsilon}{\partial F}$ terms. The velocity of the ground truth model is the same as the H-K model (Table 1, red line). The meanings of the bars are the same as in Fig. 5. The amplitude of ground-truth azimuthal anisotropy is set to be 5 per cent, and the direction of the ground-truth anisotropy varies at each layer (blue bars). Only the azimuthal anisotropy parameters are inverted for in the tests, and the ground-truth velocity model is used for kernel computation. The green bars in (b) and (c) denote the inverted azimuthal anisotropy models using kernels with and without $\frac{\partial \epsilon}{\partial F}$ term, respectively. No noise was added to the synthetic data.

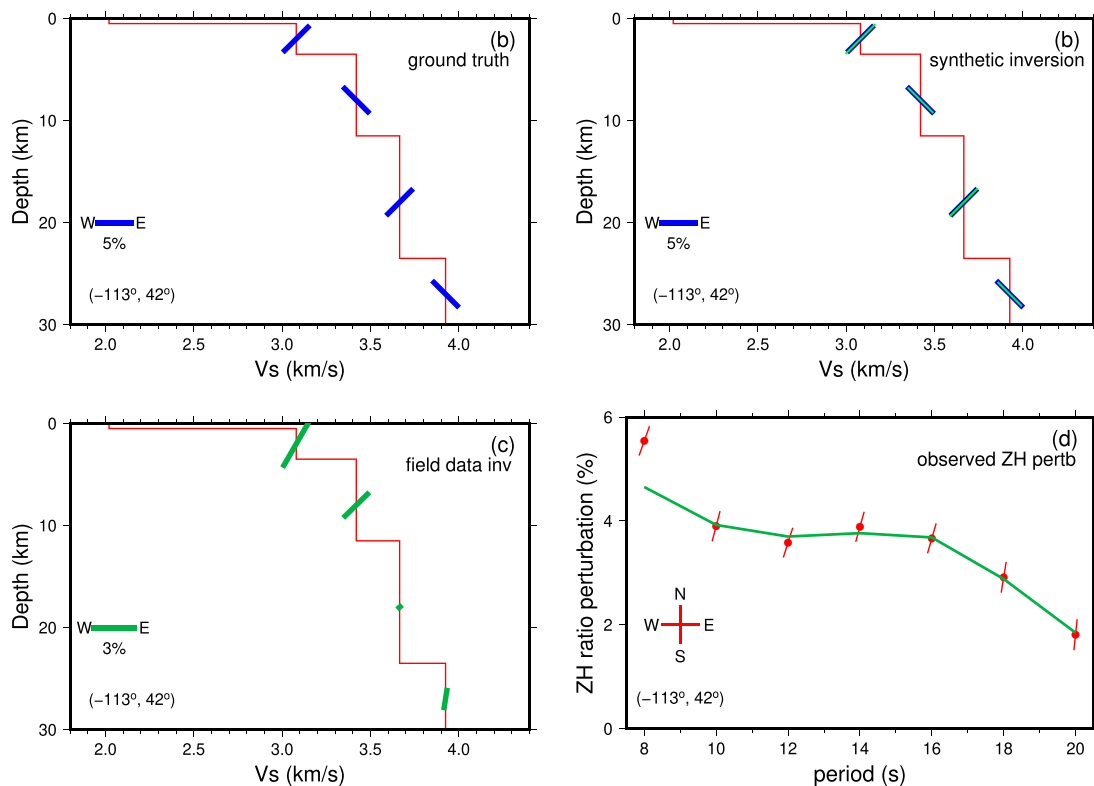


Figure 7. (a) and (b) Synthetic test and (c) and (d) application of derived formulation to invert observed Rayleigh wave ellipticity for the crustal anisotropic model located at $(-113^\circ, 42^\circ)$ in western America. The red lines are the reference model obtained from Schmandt et al. (2015). The reference model is used for kernel computation. Meanings of the bars are the same as in Fig. 5. The blue bars in (a) and (b) are the ground-truth depth-dependent azimuthal anisotropic parameters. Note that the anisotropy in top sedimentary layer is not shown, but is fixed to be the same as the second layer (a)–(c). The green bars in (b) are the synthetic inversion results of the ground-truth anisotropy model. The green bars in (c) are the inversion results of the field data in (d). Note that in (c), the amplitude of the anisotropy in the first two layers is multiplied by 0.2 for visualization. The red dots and short red bars in (d) are the observed Rayleigh wave ellipticity perturbation. The green line in (d) is the predicted ZH ratio perturbation of the inverted anisotropy model in (c). In (d), the amplitude and azimuth of the observed anisotropy are described by the location of the red dots and the direction of the red bars, respectively; the length of the red bars are set to be equal.

According to the depth-sensitivity kernel of ZH ratio and synthetic tests (Figs 7a and b) similar as in Fig. 5(c), we only inverted for the anisotropic parameters down to the depth of 30 km. Note that because the thickness of the sedimentary layer is small (0.5 km) compared with other layers (≥ 3.0 km), the top sedimentary layer was not included in the azimuthal anisotropy inversion independently. Instead, we simply assumed the anisotropic parameters in the top sedimentary layer is the same as in the second layer.

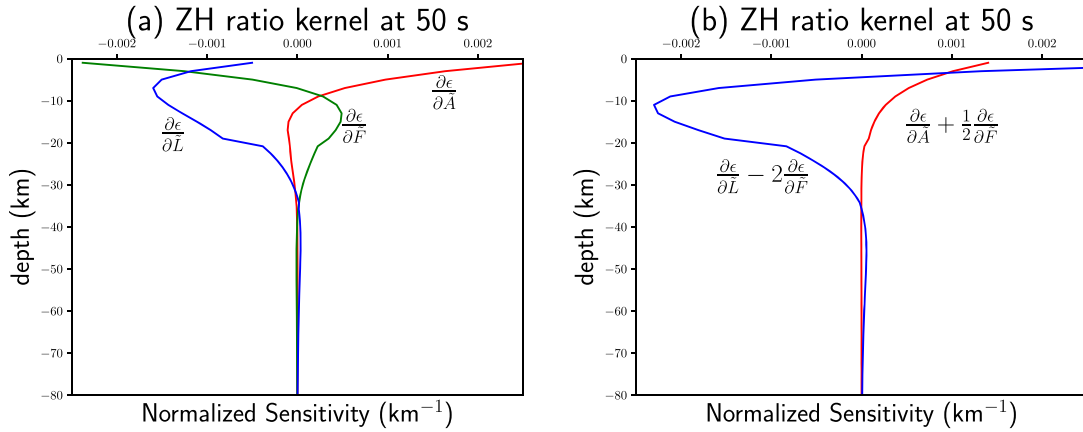


Figure 8. Depth-sensitivity kernel of ZH ratio in a reference VTI medium at 50 s. All lines are labelled. The reference model is the *IASP91* model down to the depth of 80 km. The depth-sensitivity kernels are calculated using the CPS package with the earth flattening transformation. The meanings of the colors are the same as in Fig. 4.

The inversion results and data fitting show that the derived formulation can be successfully used to invert the period-dependent Rayleigh wave ellipticity to depth for azimuthal anisotropic parameters (Fig. 7c). The inverted model shows the anisotropy dominates at the topmost 3 km with approximately 15 per cent azimuthal anisotropy, but drops to approximately 3 per cent for crust below 5 km, which is consistent with the inference of Lin & Schmandt (2014). Another interesting feature of our inverted model is that the azimuthal anisotropy is rather weak in the depths 11.5–23.5 km. This simple application indicates that Rayleigh wave ellipticity provides independent constraint on layered azimuthal anisotropy. Therefore, as shown in the synthetic tests (Fig. 5), the incorporation of backazimuth-dependent Rayleigh wave ellipticity with other types of data will potentially improve the depth resolution of azimuthal anisotropy inversion.

4.3 Backazimuth-dependent Rayleigh wave ellipticity in the longer periods

It is noteworthy that all the above derivation and discussion focus on Rayleigh wave ellipticity in a flat earth in the periods 8–20 s. In this period band, the Rayleigh wave ellipticity extracted from ambient noise cross-correlation functions seems to have a relatively high signal-to-noise ratio (Hu et al. 2020), and is observed to be backazimuth-dependent in field data (Lin & Schmandt 2014). From a theoretical point of view, the derivation and formulation are directly applicable to backazimuth-dependent Rayleigh wave ellipticity at ~ 50 s in a flat earth, or in spherical earth with the earth flattening transformation (Alterman et al. 1961; Biswas 1972).

We calculated the depth-sensitivity kernels of backazimuth-dependent ZH ratio at the period of 50 s according to eqs (50) and (51) (Fig. 8). The reference velocity was chosen to be the *IASP91* model (Kennett & Engdahl 1991) down to the depth of 80 km. The corresponding eigenfunctions of the reference model were calculated using the CPS packages with the earth flattening transformation. Numerical simulations indicate that the backazimuth-dependent Rayleigh wave ellipticity at the period of 50 s is sensitive to the anisotropic parameters in the depths down to 40 km (Fig. 8), therefore may provide constraints on anisotropic structures in the crust and upper mantle.

For even longer periods (> 50 s), the sphericity of the earth needs to be taken into consideration accurately (Tanimoto & Tsuboi 2009). On the one hand, the relation between the Backus's parametrization and the elastic coefficients C_{mn} in spherical earth (similar to eqs 20–32) has been derived before (Park 1993). Thus, the relations in eqs (43) and (44) are still valid, and the $\frac{\partial \epsilon}{\partial F}$ term still needs to be included in kernel computation in an HTI medium in the spherical coordinate. On the other hand, although the energy function (the F function in eq. 10) for an isotropic medium in spherical earth was derived in Tanimoto & Tsuboi (2009), the HTI and VTI cases (similar to eqs 11 and A12) in spherical earth need to be carefully analysed in order to determine whether eqs (50) and (51) are still valid in the spherical coordinate. In addition, the observation of backazimuth-dependent Rayleigh wave ellipticity in such long periods seem still lacking in the literature. The formal theoretical and numerical analyses for backazimuth-dependent Rayleigh wave ellipticity in spherical earth, and related work aiming at observing the backazimuth-dependent Rayleigh wave ellipticity in longer periods (> 50 s) will be highly interesting, but are not included in the current study.

5 CONCLUSIONS

In summary, we derive the formulation to relate the period-dependent azimuthal Rayleigh wave ellipticity with depth-dependent azimuthal anisotropy in a slightly horizontally transverse anisotropic medium. The derived formulation is validated by the comparison with results of the anisotropic propagator matrix method. The synthetic tests indicate that the Rayleigh wave ellipticity provides an independent and complementary constraint on Earth azimuthal anisotropic structure in addition to phase velocity dispersion. Finally, the simple application to the field data in North America shows that our formulation can be used to constrain depth-dependent azimuthal parameters.

ACKNOWLEDGEMENTS

We thank Robert Herrmann for providing the *CPS* package to compute surface wave eigenfunctions in a VTI medium, Jeffrey Park for providing the *aniprop* package to compute surface waves eigenfunctions in multilayered anisotropic structure, Fan-Chi Lin for providing the observed backazimuth-dependent Rayleigh wave ellipticity and Brandon Schmandt for providing the shear wave velocity model in North America (<https://ds.iris.edu/ds/products/emc-us-crustvs-2015/>, last accessed 2019/07/20). Comments and suggestions from two anonymous reviewers helped us to improve the manuscript greatly. This study was supported by National Natural Science Foundation of China (41904047, 41790464, 41804042, and 41874055), the China Seismic Experiment Site Project, China Earthquake Administration (2018CSES0101), and Strategic Priority Research Program of the Chinese Academy of Sciences (XDB18010304)

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APPENDIX A: PARTIAL DERIVATIVE OF ZH RATIO IN A VTI MEDIUM

The elastic coefficients C_{mn} for a VTI medium are usually represented by the Love notation ($\tilde{A}$, $\tilde{C}$, $\tilde{F}$, $\tilde{L}$, $\tilde{N}$). Here, we use the variation principle of Tsuboi & Saito (1983) and Tanimoto & Tsuboi (2009) to derive the partial derivative of the ZH ratio in a VTI medium with respect to model parameters as follows.

The $P-SV$ motion of the first-order ordinary differential wave equations of eqs (8) in a VTI medium leads to

$$\frac{\partial y_1}{\partial x_3} = \frac{1}{\tilde{C}}(y_2 + k\tilde{F}y_3), \quad (\text{A1})$$

$$\frac{\partial y_2}{\partial x_3} = -\omega^2 \rho y_1 + ky_4, \quad (\text{A2})$$

$$\frac{\partial y_3}{\partial x_3} = -ky_1 + \frac{1}{\tilde{L}}y_4, \quad (\text{A3})$$

$$\frac{\partial y_4}{\partial x_3} = -\frac{k\tilde{F}}{\tilde{C}}y_2 + \left[k^2 \left(\tilde{A} - \frac{\tilde{F}^2}{\tilde{C}} \right) - \rho\omega^2 \right] y_3 \quad (\text{A4})$$

The boundary conditions are

$$y_{1,2,3,4} \text{ are continuous at interfaces,} \quad (\text{A5})$$

$$y_2(0) = y_4(0) = 0, \quad (\text{A6})$$

$$y_{1,2,3,4}(-\infty) \rightarrow 0. \quad (\text{A7})$$

Therefore, assuming two sets of solutions y_i and z_i , of which y_i satisfies the boundary condition while z_i only satisfies the radiation boundary condition in the half-space,

$$\dot{z}_1 y_2 = \tilde{C} \dot{z}_1 \left(\dot{y}_1 - \frac{\tilde{F}}{\tilde{C}} k y_3 \right), \quad (\text{A8})$$

$$z_1 \dot{y}_2 = z_1 [(-\rho\omega^2 + \tilde{L}k^2)y_1 + k\tilde{L}\dot{y}_3], \quad (\text{A9})$$

$$\dot{z}_3 y_4 = \tilde{L} \dot{z}_3 (\dot{y}_3 + ky_1), \quad (\text{A10})$$

$$z_3 \dot{y}_4 = z_3 [-\tilde{F}k\dot{y}_1 + (k^2\tilde{A} - \rho\omega^2)y_3] \quad (\text{A11})$$

The F function can be obtained as,

$$\begin{aligned} F(\mathbf{z}, \mathbf{y}, \tilde{A}, \tilde{C}, \tilde{L}, \tilde{F}, \omega) &\equiv \frac{d}{dz}(z_1 y_2 + z_3 y_4) \\ &= \tilde{A}k^2 z_3 y_3 + \tilde{C} \dot{z}_1 \dot{y}_1 + \tilde{L}(kz_1 + \dot{z}_3)(ky_1 + \dot{y}_3) - \tilde{F}(k\dot{z}_1 y_3 + kz_3 \dot{y}_1) \\ &\quad - \rho\omega^2(z_1 y_1 + z_3 y_3). \end{aligned} \quad (\text{A12})$$

Integrating from $x_3 = -\infty$ to 0, we get

$$[z_1 y_2 + z_3 y_4]_{-\infty}^0 = \int_{-\infty}^0 F(\mathbf{z}, \mathbf{y}, \tilde{A}, \tilde{C}, \tilde{L}, \tilde{F}) dx_3. \quad (\text{A13})$$

Using the symmetry $F(\mathbf{z}, \mathbf{y}) = F(\mathbf{y}, \mathbf{z})$ and $y_2(0) = y_4(0) = 0$,

$$y_1(0)z_2(0) + y_3(0)z_4(0) = 0. \quad (\text{A14})$$

Accordingly, ZH ratio is defined as ϵ ,

$$\frac{z_4(0)}{z_2(0)} = -\frac{y_1(0)}{y_3(0)} = -\epsilon. \quad (\text{A15})$$

With the perturbation to eq. (A13), that is, $\tilde{A} \rightarrow \tilde{A} + \delta\tilde{A}$, $\tilde{F} \rightarrow \tilde{F} + \delta\tilde{F}$, $\tilde{L} \rightarrow \tilde{L} + \delta\tilde{L}$, and $\mathbf{z} \rightarrow \mathbf{z} + \delta\mathbf{z}$, $\mathbf{y} \rightarrow \mathbf{y} + \delta\mathbf{y}$, we arrive at,

$$\begin{aligned} &\int_{-\infty}^0 F(\delta\mathbf{z}, \mathbf{y}) dx_3 + \int_{-\infty}^0 F(\mathbf{z}, \delta\mathbf{y}) dx_3 + \delta k \int_{-\infty}^0 \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial k} dx_3 \\ &+ \int_{-\infty}^0 \left[\frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{A}} \delta\tilde{A} + \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{L}} \delta\tilde{L} + \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{F}} \delta\tilde{F} \right] dx_3 = 0. \end{aligned} \quad (\text{A16})$$

The symmetry of F also guarantees

$$\int_{-\infty}^0 F(\delta\mathbf{z}, \mathbf{y}) dx_3 = \int_{-\infty}^0 F(\mathbf{z}, \delta\mathbf{y}) dx_3 = [\delta y_1 z_2 + \delta y_3 z_4]_{x_3=0}. \quad (\text{A17})$$

Together with eqs (A16) and (A17), we obtain

$$\begin{aligned} \delta y_1(0)z_2(0) + \delta y_3(0)z_4(0) &= -\delta k \int_{-\infty}^0 \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial k} dx_3 \\ &- \int_{-\infty}^0 \left[\frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{A}} \delta \tilde{A} + \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{L}} \delta \tilde{L} + \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{F}} \delta \tilde{F} \right] dx_3. \end{aligned} \quad (\text{A18})$$

Similar to the derivation in Tsuboi & Saito (1983) and Tanimoto & Tsuboi (2009), due to the freedom of choice of normalization of vectors $\mathbf{z}$ and $\mathbf{y}$, it is possible to use the normalizations $y_3(0) = 1$ and $z_2(0) = 1$. The left-hand side of eq. (A18) becomes $\delta y_1(0) = \delta \epsilon$. Then,

$$\delta \epsilon = -\delta k J_{zy} - \int_{-\infty}^0 \left[\frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{A}} \delta \tilde{A} + \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{L}} \delta \tilde{L} + \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial \tilde{F}} \delta \tilde{F} \right] dx_3, \quad (\text{A19})$$

where J_{zy} is defined as

$$J_{zy} = \int_{-\infty}^0 \frac{\partial F(\mathbf{z}, \mathbf{y})}{\partial k} dx_3. \quad (\text{A20})$$

By considering $\mathbf{z} = \mathbf{y}$, eq. (A18) leads to

$$\delta k = -\frac{1}{J_{yy}} \int_{-\infty}^0 \left[\frac{\partial F(\mathbf{y}, \mathbf{y})}{\partial \tilde{A}} \delta \tilde{A} + \frac{\partial F(\mathbf{y}, \mathbf{y})}{\partial \tilde{L}} \delta \tilde{L} + \frac{\partial F(\mathbf{y}, \mathbf{y})}{\partial \tilde{F}} \delta \tilde{F} \right] dx_3. \quad (\text{A21})$$

Substituting eq. (A21) in eq. (A19), we have

$$\delta \epsilon = \int_{-\infty}^0 [K_{\tilde{A}} \delta \tilde{A} + K_{\tilde{L}} \delta \tilde{L} + K_{\tilde{F}} \delta \tilde{F}] dx_3, \quad (\text{A22})$$

where $K_{\tilde{A}}$, $K_{\tilde{L}}$ and $K_{\tilde{F}}$ are the analytical partial derivatives of ZH ratio with respect to $\tilde{A}$, $\tilde{L}$ and $\tilde{F}$, respectively; and they can be calculated as

$$K_{\tilde{A}} = \frac{J_{zy}}{J_{yy}} k^2 y_3^2 - k^2 z_3 y_3, \quad (\text{A23})$$

$$K_{\tilde{L}} = \frac{J_{zy}}{J_{yy}} (k y_1 + \dot{y}_3)^2 - (k z_1 + \dot{z}_3)(k y_1 + \dot{y}_3), \quad (\text{A24})$$

$$K_{\tilde{F}} = -2k \frac{J_{zy}}{J_{yy}} \dot{y}_1 y_3 + k(\dot{z}_1 y_3 + z_3 \dot{y}_1). \quad (\text{A25})$$

The above formulation can be reduced to the results of Tsuboi & Saito (1983) and Tanimoto & Tsuboi (2009) in the isotropic case.